

SUPPLY CHAIN DELIVERY PERFORMANCE REPORT

End-to-End Order Fulfillment Analysis & Predictive Insight

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1. Executive Summary

This report presents a comprehensive analysis of the delivery operations of a global e-commerce company managing end-to-end order fulfillment across multiple regions. The analysis covers 172,765 orders spanning January 2015 through January 2018, focusing on identifying root causes of chronic late deliveries, quantifying their financial impact, and establishing a data-driven framework for improvement.

The headline finding is stark: 54.71% of all orders are delivered late, costing the business \$2.1M in eroded profitability on delayed orders alone. While total profit across fulfilled orders reached \$7.5M, the persistent late-delivery problem represents both a significant financial drain and a major customer experience risk. A Random Forest predictive model achieved 74% accuracy in identifying at-risk orders before they become late.

Key Finding	Metric
Overall Late Delivery Rate	54.71%
On-Time Delivery Rate	45.29%
Total Orders Analyzed	172,765
Total Profit (profitable orders)	\$7.5M
Profit at Risk (delayed orders)	\$2.1M
Predictive Model Accuracy (RF)	74%

2. Business Context & Objectives

The company operates a global e-commerce platform selling products across categories including sporting goods, fitness equipment, outdoor gear, footwear, and apparel across multiple international regions.

Core Business Problem

Actual shipping times frequently deviate from scheduled delivery windows, creating eroded customer trust, reduced order profitability, and an inability to make reliable commitments to buyers at point of purchase.

Analytical Objectives

- Understand the current state of delivery performance across all dimensions (region, mode, time, segment)
- Quantify the financial impact of delays on order profitability
- Identify the primary operational bottlenecks driving late deliveries
- Build a predictive model to flag high-risk orders before they are shipped
- Deliver actionable recommendations to improve on-time delivery and profitability

3. Key Performance Indicators (KPIs)

The following KPIs were computed directly from the cleaned dataset to establish a performance baseline.

172,765 Total Orders	94,523 Late Deliveries	54.71% Late Delivery %	45.29% On-Time Delivery %
\$7.5M	\$2.1M	3 days	\$22.03

Total Profit (profitable orders)	Profit at Risk (delayed orders)	90th Percentile Delay	Avg Order Profit
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• Late Delivery Rate at 54.71%	More than half of all orders arrive late. This is not an edge-case problem, it is the default experience for the majority of customers.
• \$2.1M Profit at Risk	Orders that experienced delays collectively generated \$2.1M in profit that is under constant pressure from further operational deterioration.
• 90th Percentile Delay = 3 Days	Even the most extreme cases are contained within 3 days of lateness, suggesting the problem is systemic and process-driven rather than catastrophic.

4. Profitability Analysis

4.1 Profitability Distribution

Order-level profitability was classified into three tiers based on Order Profit Per Order. While 80.7% of orders are profitable, the 18.7% loss-making share represents a meaningful drag that is disproportionately concentrated among delayed shipments.

Profitability Distribution (%)

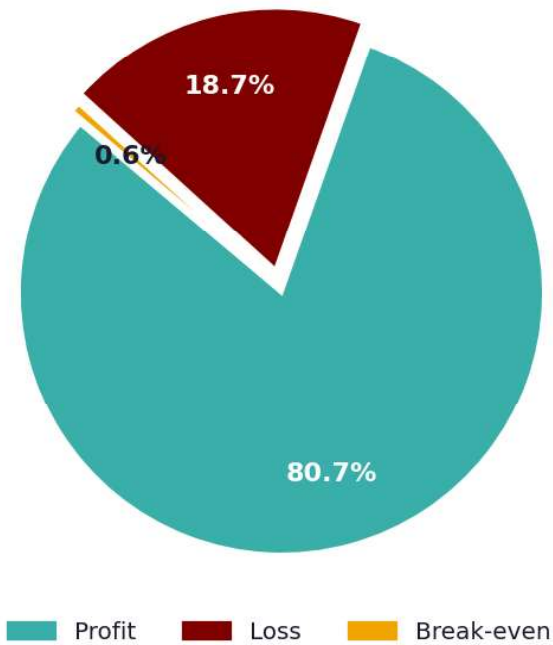


Figure 1: Profitability Distribution — 80.7% Profit | 18.7% Loss | 0.6% Break-even

Tier	Definition	Share
Profit	Order Profit Per Order > 0	80.7%
Loss	Order Profit Per Order < 0	18.7%

Break-even

Order Profit Per Order = 0

0.6%

4.2 Delay Distribution & Profit vs. Delay Days

The delay distribution shows that 31.0% of all orders arrive exactly 1 day late the single largest cohort. Combined, orders delayed by 1–4 days account for 54.7% of all order volume, directly mapping to the overall late delivery rate.



Figure 2: Delay Distribution (left) and Profit Analysis by Delay Days (right)

A critical observation: mean profit per order remains stable at approximately \$20–\$23 across all delay levels. The profitability problem is therefore driven by volume of delayed orders, not by individual order economics making systemic throughput improvement the highest-leverage intervention.

5. Bottleneck Detection

Delay percentage was computed across six key operational dimensions. The charts below reveal where the fulfilment process breaks down most severely.

